

Experimental and theoretical investigations of affect in language and cognition

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Day 2: Psychology & Emotion Theory

<https://mcmoyer11.github.io/pages/essli-2026.html>

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Course Outline

~~Day 1: intro / baby r~~

Day 2: psych

Day 3: psycho-ling

Day 4: sem-prag

Day 5: decision making

What is emotion?

- **I am furious** = emotion label.
- **murder** = negative valence, high arousal, but not an emotion.
- **This policy is disgusting** = affective evaluation
- **I feel uneasy** = affective state, not necessarily a discrete emotion
- **I've been irritable all day** → **mood**: more diffuse and longer-lasting; may lack a clear object
 - The emotion/mood distinction is not perfectly categorical
 - duration and identifiable cause are common distinctions (Beedie et al., 2005)
- What about: **I feel tired/lazy/stupid?**
 - Lay conceptualisations of feelings?

Are we describing the stimulus, the person's state, or their interpretation?

Affective meaning in language ≠ inferred emotion ≠ experienced emotion

See: Scherer & Moors (2019)

What does emotion involve?

- **Appraisal:** what does this event mean for me?
 - Unpleasant? Relevant? Goal-congruent? Who caused it? Can I cope/control?
- **Physiological change:** autonomic/bodily responses.
 - Sweating, heart rate, etc. which may help you get ready to act
- **Action tendency:** How should I react?
 - Approach, avoid, attack, withdraw...
- **Expression:** face, voice, posture, gesture
 - Helps communicate information to others
- **Subjective feeling:** conscious experience
- **Labelling:** Assigning a label or category
 - “I am angry”

These components covary imperfectly.

See Scherer & Moors (2019)

Detecting, eliciting & regulating emotion

Computational work often asks whether a text *contains* an emotion category. Human research asks several distinct questions:

- Does the person recognise the intended emotion?
- Does the stimulus elicit an emotional response?
- Does it alter attention, interpretation or memory?
- Does the person regulate or suppress that response?
- Does it affect judgement or behaviour?

These are different outcomes and require different measures.

Consider the difference:

meaning encoded in the stimulus

meaning inferred by the comprehender

response elicited by that interpretation

Competing questions

- **Discrete/basic emotion theories**
→ Are there qualitatively distinct emotion systems?
- **Dimensional theories**
→ Can affective states be located in a continuous space?
- **Appraisal theories**
→ Why does the *same event* produce different emotions?
- **Constructionist theories**
→ How do bodily signals, context and conceptual knowledge become an emotion category?



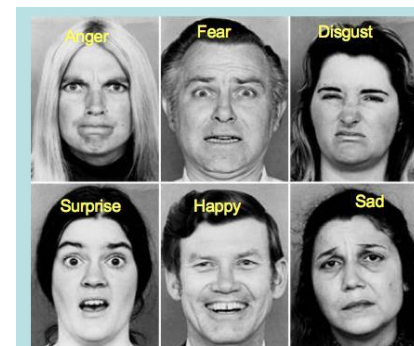
Discrete emotions: Ekman

Ekman's claim:

- Emotions are evolved, discrete systems

Predictions:

- Distinct, recognisable emotional categories
- Evolved to deal with specific adaptive problems
- There are cross-cultural regularities
- See: Ekman, 1992; Cordaro et al., 2020



Discrete emotions: Universal?

- If emotional expressions are culturally learned, why do people who have never seen them produce them?
- **Ekman & Friesen (1971):** Consistent association between facial expression and discrete emotion tested in Papua New Guinea, with no Western media influence
- **Matsumoto & Willingham (2009):** Congenitally blind athletes show the same spontaneous facial expressions after winning and losing as sighted athletes
- **Tracy & Matsumoto (2008):** Congenitally blind athletes showed the characteristic pride display after winning (extending to bodily expressions)

Discrete emotions: Ekman

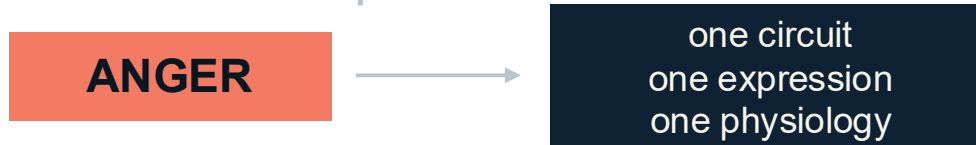
- Discrete emotions (anger, fear, sadness, etc.) are **qualitatively different** evolved emotion systems:
 - distinctive signals and physiology
 - automatic appraisal
 - rapid onset
 - distinctive subjective experience.
- Expression is biologically prepared (**but can be modified by learning and culture**)

But what exactly is universal?

- Cross-cultural agreement $\neq$ one-to-one mapping between **face and internal emotion**
- Recognition depends on **task format, available labels and context**
- Free-label/categorisation studies show substantially more cultural variability

Problems with emotion theories:

The classical assumption:



The empirical evidence:



Appraisal theory:

Consider how evaluations of events and context shape emotional experience through a series of **evaluations**

Constructed emotion theories:

Consider how bodily signals, context and conceptual knowledge are combined to **construct** emotion categories

Appraisal theories

Stimuli / events are evaluated on a series of evaluative dimensions. For example:

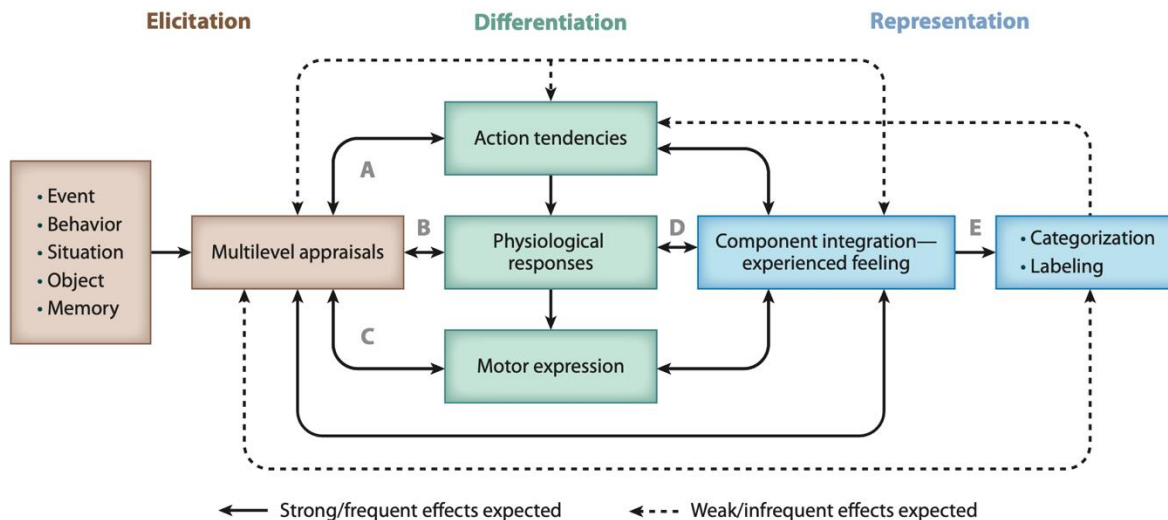
- **Relevance:** Does this matter to my goals?
- **Goal congruence:** Does it help or obstruct them?
- **Agency:** Who caused it?
- **Control / coping potential:** Can I do anything about it?
- **Norm compatibility:** Is it consistent with expectations/values?

Different appraisal patterns are proposed to help **differentiate emotional responses**

See Scherer & Moors (2019)

Appraisal Theories

- Extends beyond valence
- Appraisals help shape action tendencies, physiological responses and expression
- These changes contribute to experienced feeling
- Feelings may then be categorised / labelled
- Feedback between components makes the process **dynamic**



Someone blocks an important goal → negative goal-incongruent another person is responsible
relatively high certainty
some potential to act or confront
= ANGER

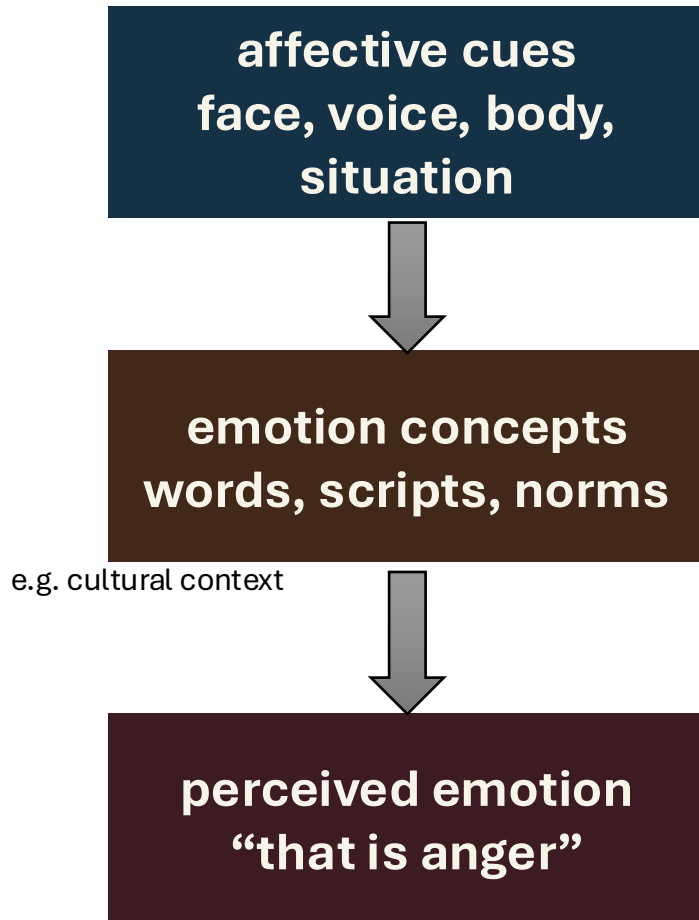
negative, goal-incongruent
•threatening outcome
•greater uncertainty
•lower perceived control
FEAR

Multicomponent process model by Scherer and Moors (2019)

Constructed Emotions

- An emotion category such as **fear** is **not one biological state**.
- Different instances of fear can involve quite different bodily responses, actions and neural patterns.
- What makes them instances of *fear* is that the brain categorises them as serving a similar situated function.
- **Emotion = situated categorisation of sensory information using prior experience and conceptual knowledge.**
- See: Feldman Barrett (2006, 2017)

Constructing emotions: Feldman Barret



Some key terms:

- **Allostasis** = anticipatory regulation of the body's physiological resources.
- **Interoception** = prediction of the body's internal physiological state.
- **Conceptualisation** = using learned, situation-sensitive conceptual knowledge to make sensory information meaningful.

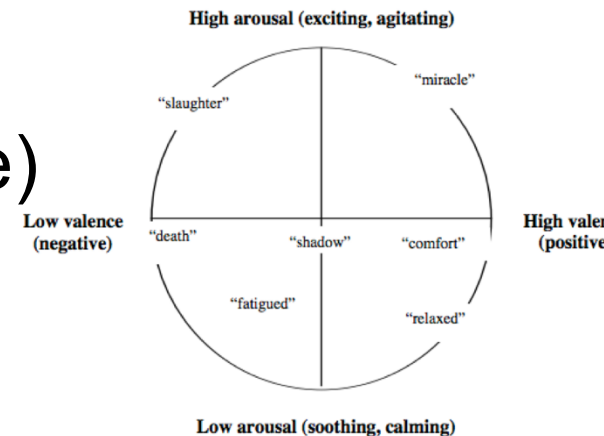
Predictions about bodily **consequences + expectations** based on past experience = emotion and subsequent action

Language constructs emotion

- **Has an active role** – not just the output system
- Emotion words can therefore help **constitute the categories through which affective information is perceived**, rather than simply naming an independently represented emotion afterwards (Lindquist et al., 2015)
- **Consider:** anger – irritation – frustration – resentment – indignation
- **What is the difference?**

Dimensional Approaches

- Emotions are represented as a combination of some continuous variables:
- **VALENCE** (pleasant vs. unpleasant)
- **AROUSAL** (calming/sleepy vs. exciting/activated)
- Dominance (high vs. low control)
- Also: Approach vs. avoid (and more)
- **Very easy to operationalize →**



Osgood et al. (1957): Semantic differential (EPA)

Mehrabian & Russell (1974): From semantic to affective dimensions (PAD)

Russel (1980): Affect has a geometric structure

Bradley and Lang (1994): Affect becomes easier to measure (SAM)

Emotion norms: Stimuli at last!

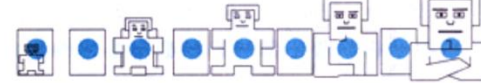
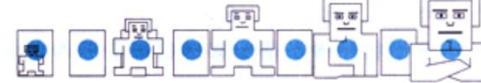
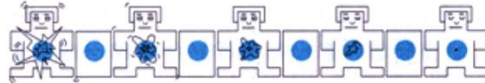
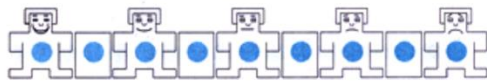
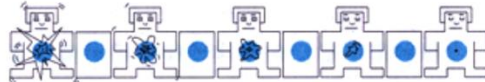
- Dimensional approaches provide neat stimuli:
- Bradley & Lang (1999): Affective Norms for English Words
- Warriner et al. (2013): Norms of valence, arousal, and dominance for 13,915 English lemmas
- Mohammad (2018): Obtaining reliable human ratings of valence, arousal, and dominance for 20,000 English words
 - And many others!
- We return to issues with these throughout the course!

Collecting Affective Norms

- **Bradley & Lang (1999)**: affective norms for ~1,000 English words
- Participants silently read each word on **valence, arousal and dominance**
- Each participant rated 100–150 words at own pace
- Ratings were made using the **Self-Assessment Manikin (SAM)**: 9 possible positions on each pictorial scale
 - Different norms use different methods
 - Translation attempts must use the same method to be comparable across languages (e.g., ANGST; Schmidtke et al., 2014)



Name (Please Print)	
Home Phone	Work Phone



Affective Norms for English Words. All Subjects Bradley, M.M., & Lang, P.J. (1999)

Description	Word No.	Valence Mean(SD)	Arousal Mean(SD)	Dominance Mean (SD)	Word Frequency
dentist	589	4.02 (2.23)	5.73 (2.13)	3.80 (2.16)	12
depressed	107	1.83 (1.42)	4.72 (2.95)	2.74 (2.13)	11
depression	108	1.85 (1.67)	4.54 (3.19)	2.91 (2.27)	24
derelict	722	4.28 (1.84)	4.10 (1.94)	4.78 (1.56)	1
deserter	109	2.45 (1.80)	5.50 (2.55)	3.77 (2.29)	.
desire	508	7.69 (1.39)	7.35 (1.76)	6.49 (1.83)	79
despairing	110	2.43 (1.47)	5.68 (2.37)	3.43 (2.11)	4
despise	111	2.03 (1.38)	6.28 (2.43)	4.72 (2.80)	7
destroy	112	2.64 (2.03)	6.83 (2.38)	4.94 (2.86)	48
destruction	723	3.16 (2.44)	5.82 (2.71)	3.93 (2.29)	38
detached	113	3.86 (1.88)	4.26 (2.57)	3.63 (2.15)	12
detail	724	5.55 (1.58)	4.10 (2.24)	5.21 (1.60)	72

Excerpt of instructions:

At one extreme of this scale, you are happy, pleased, satisfied, contented, hopeful. When you feel completely happy you should indicate this by bubbling in the figure at the left... The figures also allow you to describe intermediate feelings of pleasure, by bubbling in any of the other pictures. If you feel completely neutral, neither happy nor sad, bubble in the figure in the middle (point to middle SAM figure).

Collecting Affective Norms

- **Warriner et al. (2013):** 13,915 English lemmas
 - compiled from three sources: Bradley and Lang's (1999), Van Overschelde et al. (2004), and SUBTLEX-US (Brysbaert & New, 2009).
- Words were rated on one dimension per participant (different participants rated valence, arousal, dominance)
- Each participant rated ~350 words
- US residents on mechanical turk
 - globally representative???
- Ratings were on **9-point numerical scales** without the SAM pictures
 - The scale ranges from 1 (happy [excited; controlled]) to 9 (unhappy [calm; in control]).

Collecting Affective Norms

- **Mohammad (2018): NRC-VAD: ~20,000 English terms**
- **Participants saw 4 words at a time**
- **Selected the word with the highest and lowest valence, arousal or dominance**
 - A relative judgement
- **No numerical rating scale was required**
- **Repeated comparisons were aggregated into continuous 0 to 1 VAD scores**

Q1. Which of the four words below is associated with the MOST happiness / pleasure / positiveness / satisfaction / contentedness / hopefulness OR LEAST unhappiness / annoyance / negativity / dissatisfaction / melancholy / despair?
(Four words listed as options.)

Q2. Which of the four words below is associated with the LEAST happiness / pleasure / positiveness / satisfaction / contentedness / hopefulness OR MOST unhappiness / annoyance / negativity / dissatisfaction / melancholy / despair?
(Four words listed as options.)

Stimulus-level confounds

Aim to match and control for as many as possible

- frequency
- length
- concreteness and imageability
- familiarity
- semantic category
- social offensiveness
- valence extremity
- arousal (if looking at valence alone)
- cultural and language-specific associations

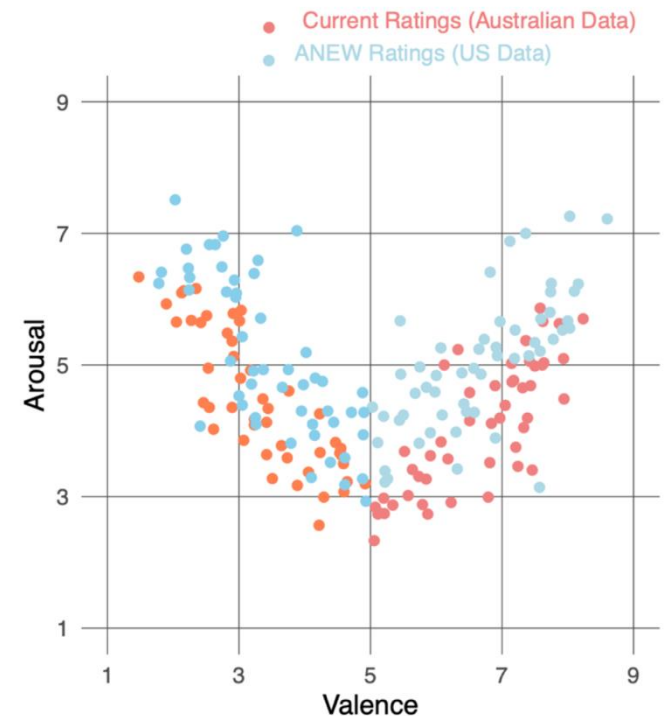
Emotion Norms: Issues

- Drift over time: arousal
- Australian participants in 2021

Details here:

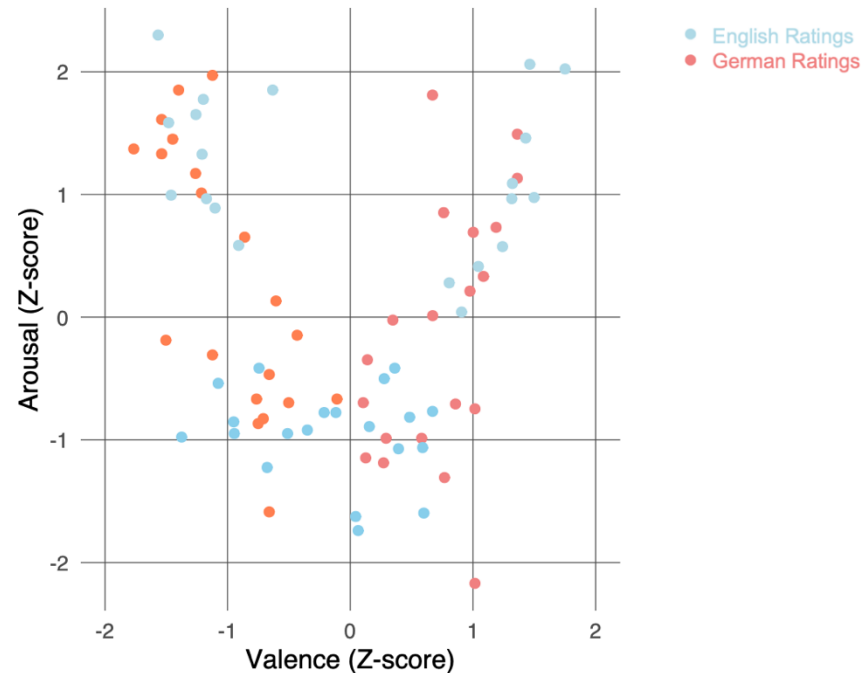


Norms are population- and time-specific estimates, not fixed lexical properties



Emotion Norms: Issues

- Across languages
- How similar are these:



Does translation preserve affective meaning?

Individual-level confounds

- Attention
- Motivation
- Alexithymia
- Autism Spectrum traits
 - difficulty recognising and understanding emotion
- Bilingualism – language experience

Case Study:

The Foreign Language Effect

Bilinguals make different decisions when responding in their foreign language:

- Decreased Loss Aversion (Keysar et al., 2012)
- Reduced Causality bias (Diaz-Lago & Matute, 2018)
- Less superstitious (Hadjichristidis et al., 2017)
- Reduces 'winning streak' beliefs (Gao et al., 2015)
- More eco-friendly consumption (less disgust; Geipel et al., 2018)
- Reduced mental imagery (Hayakawa & Keysar, 2018)
- More sacrificial responding in moral dilemmas...

Bilingual emotion processing

- **Later-learned languages are less connected to affective, autobiographical experiences** than languages acquired through early emotionally rich interaction (Ferré et al., 2018; Imbault et al., 2021)
 - Pronounced for Taboo words (Dewaele, 2004) and terms of endearment (Dewaele, 2008)
 - Subjective and objective measures don't always align (Thoma, 2024)
- Emotional “impact” varies with experience (it is not a virtue of L2)
- Consequently, language can affect **downstream judgement and decision-making** (e.g., foreign-language effect)

Case study: The Foreign Language Effect

- How do we know the stimulus has the “correct” emotion or level of emotion encoded?
- How can we control for cultural differences?
- How can we know affect/emotion is responsible for the changes in decision-making?
- **Can we separate cognition and emotion?**

Measure $\neq$ construct

- We measure **components / consequences**:
- **Experience**
self-report, ratings
- **Physiology**
heart rate, EDA, pupil dilation, EEG/fMRI
- **Expression**
face, voice, posture
- **Behaviour**
choice, approach/avoidance
- **Processing**
RT, eye movements, memory, attention

Emotion Measures:

Introduction to the issues

- Self-reported valence measures conscious evaluation
- Arousal ratings do not necessarily indicate physiological arousal.
- Reaction times index processing consequences, not emotion directly
- Priming effects can reflect semantic association, response preparation or attention
- Lexical decisions measure word recognition under particular conditions
- Physiological measures are not one-to-one “readouts” of particular emotions
- Downstream judgements involve many processes

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